INFINITE TETRADS OF CONGRUENT FACTORIALS

OCTAVIO A. AGUSTÍN-AQUINO AND JOSÉ HERNÁNDEZ SANTIAGO

ABSTRACT. We prove that there exist infinitely many primes p such that there exist k_1, k_2, k_3, k_4 distinct such that

$$k_1! \equiv k_2! \equiv k_3! \equiv k_4! \equiv 1 \pmod{p}.$$

1. INTRODUCTION

Following [1], we write $a \backslash b$ to mean a divides b. The following fact is relatively well-known.

Lemma 1.1 (Wilson's symmetry)**.** *Let p be an odd prime and $0 \le n \le p-1$ an odd integer such that $n! \equiv 1 \pmod{p}$. Then*

$$(p-1-n)! \equiv 1 \pmod{p}.$$

Proof. By Wilson's theorem we have $(p-1)! \equiv -1 \pmod{p}$. Then

$$(p-1)! = (p-1-n)! \prod_{k=1}^{n} (p-k).$$

Thus, modulo p, we have

$$\begin{aligned}(p-1-n)! \prod_{k=1}^{n} (p-k) &\equiv (p-1-n)!(-1)^n n! \\ &\equiv -(p-1-n)! \equiv -1 \pmod{p},\end{aligned}$$

hence $(p-1-n)! \equiv 1 \pmod{p}$. □

Wilson's symmetry is crucial for the following construction.

Lemma 1.2 (Valid tetrad)**.** *Let $q \ge 3$ be an odd integer, and p a prime divisor of $q! - 1$. If p is distinct from $q+2$ and $2q+1$, then*

$$k_1 = 1, \quad k_2 = q, \quad k_3 = p-2, \quad k_4 = p-1-q,$$

are four different integers such that $k_i! \equiv 1 \pmod{p}$ for $1 \le i \le 4$.

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Proof. Since $p \backslash (q! - 1)$, then $p > q$. Otherwise, $q! \equiv 0 \pmod p$, which contradicts that $q! \equiv 1 \pmod p$. We already have $1!, q! \equiv 1 \pmod p$, and by Wilson's symmetry, $(p-2)!, (p-1-q)! \equiv 1 \pmod p$ because q is odd. If the k_i were not distinct, then $p-2 = 1$ if and only if $p = 3$, $p-1-q = 1$ if and only if $p = q+2$, $q = p-2$ if and only if $p = q+2$, or $q = p-1-q$ if and only if $p = 2q+1$, or $p-2 = p-1-q$ if and only if $q = 1$, but all of these are blocked by the hypotheses. □

2. Main Theorem

Theorem 2.1. *There exist infinitely many primes p such that*

$$|\{1 \leq n \leq p-1 : n! \equiv 1 \pmod p\}| \geq 4.$$

Proof. Let $q = 6t + 1$ for an integer $t \geq 1$. It follows that q is odd, while

$$q + 2 = 6t + 3 = 3(2t + 1) \quad \text{and} \quad 2q + 1 = 12t + 3 = 3(4t + 1),$$

which implies $q + 2$ and $2q + 1$ are composite. Thus any prime divisor p of $q! - 1$ cannot be equal to $q + 2$ or $2q + 1$. Now $q! - 1 > 1$, and hence such a prime p exists. By the valid tetrad lemma, there exist k_1, k_2, k_3, k_4, all distinct, such that their factorials coincide in having residue 1 modulo p.

Let S be a finite set of positive primes with such a property. Choose $q = 6t + 1$ larger than any of those primes. Then $q! - 1$ has a positive prime divisor p that is larger than q, and thus it cannot belong to S, and the result follows. □

Remark 2.2. This result is connected to problem A15 of Guy's compendium [2, p. 54] and the corresponding observation by Mąkowski [3].

3. Some Computations

3.1. **Euclidean Scheme.** We can obtain sequences of primes with at least a tetrad of factorials congruent to 1 using the Euclidean scheme of the proof (see [1, Section 4.3]). If we begin with $t = 1$, then we have $q = 6 \cdot 1 + 1 = 7$. But $7! - 1 = 5039$ is prime, and the smallest q of the form $6t + 1$ that is larger than 5039 is 5041. The least prime factor of $5041! - 1$ is 20549. This was found considering that if we write the prime divisor as $s = 5041 + r$, then by Wilson's theorem

$$(s-1)! \equiv -1 \pmod s.$$

By the definition of s, we have

$$(s-1)! = 5041! 5042 \cdot 5043 \cdot \cdots \cdot (s-1),$$

and modulo s we have

$$\begin{aligned} 5042 \cdot 5043 \cdot \dots \cdot (s-1) &\equiv (-r+1)(-r+2)\cdots(-1) \\ &\equiv -(r-1)! \pmod{s} \end{aligned}$$

because r is even. Thus

$$-1 \equiv 5041!(-(r-1)!) \pmod{s}.$$

Hence $5041! \equiv 1 \pmod{s}$ if, and only if, $(r-1)! \equiv 1 \pmod{s}$. A direct check using the primes greater than 5041 with a Python script yields $r = 15508$. The next q is 20551, and the script yields $r = 4296$ for the prime factor $s = 24847$. After this the algorithm takes more than a few seconds to finish the computation.

If we begin with $t = 2$, then $q = 6 \cdot 2 + 1 = 13$, and the least prime factor of $13! - 1$ is 1733. The next q of the form $6t + 1$ is 1735. After this the algorithm takes more than a few seconds to finish the computation.

If we begin with $t = 3$, then $q = 6 \cdot 3 + 1 = 19$, and the least prime factor of $19! - 1$ is 653. Now the next q of the form $6t + 1$ is 655, and the least prime factor of $655! - 1$ is 5407. After this the algorithm takes more than a few seconds to finish the computation.

3.2. **Tetrads-Only Primes.** It is interesting to calculate the sequence of primes p that admit exactly one tetrad of factorials congruent to 1 modulo p. Using a simple Python script, we can readily find its first terms:

$$17, 53, 59, 73, 83, 89, 97, 139, 239, \dots$$

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Source Code

3.3. **Euclidean Scheme.**

```
import sympy as sp

def search_prime_divisor(n):
  # Here n is assumed to be
  # an odd number.
  for s in sp.primerange(n + 1, 10**7):
    r = s - n
    prod = 1
    for u in range(1,r):
```

```
        prod = (prod * u) % s
    if prod == 1:
        return s,r
```

3.4. **Tetrads-Only Primes.**

```
import sympy as sp

for p in sp.primerange(2, 10000):
    restos = []
    fact = 1

    for n in range(1, p):
        fact = (fact * n) % p

        if fact == 1:
            restos.append(n)

            # Optional early stop:
            # we only care about exactly 4
            if len(restos) > 4:
                break

    if len(restos) == 4:
        print(p)
```

References

[1] Ronald L. Graham, Donald E. Knuth, and Oren Patashnik. *Concrete Mathematics*. 2nd ed. Boston, MA: Addison Wesley, 1994.

[2] Richard K. Guy. *Unsolved Problems in Number heory*. 3rd ed. Springer, 2004.

[3] Andrzej Mąkowski. “On a number-theoretical problem of Erdös”. In: *Elemente der Mathematik* 38 (1983), pp. 101–102.

Instituto de Física y Matemáticas, Universidad Tecnológica de la Mixteca, Huajuapan de León, Oaxaca, México

Email address: octavioalberto@mixteco.utm.mx

URL: www.utm.mx/~octavioalberto

Escuela Superior de Matemáticas No. 2, Universidad Autónoma de Guerrero, Ciudad Altamirano, Guerrero, México

Email address: 15620@uagro.mx

URL: https://euclid.uagro.mx/